

17LIVE: Offering glitchless and diverse live streaming services through cloud infrastructure and AI

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ABOUT 17LIVE

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About 17LIVE

With the highly scalable cloud infrastructure and integrated data analytics and AI tools of Google Cloud, 17LIVE provides glitchless streaming experiences while optimizing its operations.

17LIVE is a live streaming company founded in Taiwan in 2015 and has rapidly expanded its presence worldwide. Its live streaming platform offers a diverse range of content, including music, dance, comedy, gaming, V-liver, and lifestyle channels with a unique interactive experience to more than 50 million registered users from 133 countries globally.

Industries: Media & Entertainment

and the content
performance on i
video streaming i

Location: Taiwan

Google Cloud results

- Ensures glitchless live streaming despite instant traffic surges with the autoscaling feature of GKE
- Supports capacity of up to two million concurrent viewers on the live streaming platform through flexible content

Shorte
depl
by 80%

Monitoring

Google Cloud (<https://cloud.google.com/>) Cloud
Storage (<https://cloud.google.com/storage/>),

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://cloud.google.com/compute/>)

Cloud CDN (<https://cloud.google.com/cdn/>) ow/),

Cloud Composer

(<https://cloud.google.com/composer/>)

Cloud (<https://cloud.google.com/composer/>),

Monitoring (<https://cloud.google.com/monitoring/>)

(<https://cloud.google.com/products/operations/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Looker (<https://cloud.google.com/looker/>)

Operations

(<https://cloud.google.com/products/operations/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

delivery
enabled by
Cloud CDN

- Helps
improve live
streaming
content
performance
with data
insights
generated
by BigQuery
- Reduces
40% of
streaming-
related
issues
through
system error
predictions
powered by
Vertex AI

The global live video streaming market continues to bloom

(<https://www.globenewswire.com/en/news-release/2023/04/27/2656699/28124/en/Global-Live-Streaming-Market-Report-2023-Increasing-Penetration-of-Mobile-Devices-and-Internet-Users-Drives-Growth.html>)

following an explosive growth during the COVID-19 period. Not only have the numbers of users of live streaming platforms grown rapidly, but the diversity of live streaming content has also increased, ranging from live concerts, gaming, sports, virtual idols to cooking, comics, painting and education.

17LIVE (<https://about.17.live/about/>) is one of the pioneers in the live streaming industry. Founded in 2015 in Taiwan, the company entered the Japanese market in 2017 and within a year established a top position in the live streaming platform industry in Japan, Taiwan, Hong Kong and other regions. With a user base of over 50 million registered users from 133 countries globally, 17LIVE offers a diverse range of content, including music, dancing, comedy, gaming, V-liver, and lifestyle channels. Its platform provides content creators with an excellent opportunity to connect with a highly-engaged community and build a loyal fan base.

Eric Hsu, VP of Engineering at 17LIVE attributes the platform's success to its support of authentic expressions and borderless human connections. "Our goal is to offer a community-based platform where any users from any region can easily become a live streamer sharing their talents and passions, and connect with people having the same interests across the world at any time. Meanwhile, the audience can enjoy excellent viewing experiences," he notes.

At first, 17LIVE deployed its live streaming application on a public cloud platform that didn't have a data center in Taiwan. As its business in Taiwan grew, the digital media company realized that it needed to enhance the scalability capability of its app infrastructure and lower the network latency of its streaming services. In 2017, 17LIVE decided to migrate its application to Google Cloud (<https://cloud.google.com/>), because it supports highly automated scalability and offers better connectivity

in Taiwan through a local data center. Besides, as 17LIVE had already been using BigQuery (<https://cloud.google.com/bigquery>) to analyze its business data in real time for service and operation improvement, it could therefore deploy its data pipeline fully on Google Cloud and expand the applications of its data.

"We have been using BigQuery for data analytics since the launch of our live streaming platform and found no other product could offer the same level of scalability and cost-effectiveness," explains Hsu. "Moving our infrastructure to Google Cloud was a self-evident choice, because we can not only leverage its greater scalability and data center in Taiwan to improve our service quality, but also optimize our data analytics workflows on one single platform."

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Enabling high scalability and low latency for better streaming experiences

17LIVE now uses containerized microservices in

Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) as the infrastructure backbone of its live video streaming application, with some mutable services running in Compute Engine

(<https://cloud.google.com/compute>). Hsu says that the horizontal pod autoscaling (HPA) feature of GKE has given significant help, because it allows the 17LIVE team to ensure the service quality of its streaming platform with minimum effort by automatically scaling up microservices when the CPU usage surpasses a certain amount.

On top of that, by adopting the containerization technology of GKE, the 17LIVE team is able to easily create isolated environments for the development and testing of different features, which has helped the team boost its development efficiency and change its software release cycle from once every two weeks to daily. With the HPA feature, 17LIVE has also successfully shortened the time required to deploy a new software version from 30 to six minutes. This means that the users of 17LIVE's streaming platform can now enjoy more new features in a rather glitchless way.

"The autoscaling capability and developer-friendly features of GKE have greatly reduced our DevOps workloads while enhancing our streaming service quality," notes Hsu. "We can now tackle any instant traffic surges more easily and add new features much more frequently."

The comprehensive global network of Google Cloud has also enabled 17LIVE to provide better live streaming experiences. By deploying its application in the data center of Google Cloud in Taiwan, 17LIVE has reduced network latency for users in Taiwan from 300 to only 50 milliseconds. It also leverages Cloud CDN (<https://cloud.google.com/cdn>) to cache static API responses and streaming content close to its users around the world, keeping content delivery time low and making it possible to realize the Concert Mode feature on its streaming platform that can accommodate two million viewers at the same time.

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Improving user engagement through data analytics and AI predictions

17LIVE has been using BigQuery to generate data insights since 2015. After moving the infrastructure of its platform to Google Cloud, the digital media company has built a more streamlined data processing system to fully unlock the value of its business data. It now uses [Pub/Sub](https://cloud.google.com/pubsub) (<https://cloud.google.com/pubsub>) to ingest user data from its streaming platform to BigQuery for analysis, [Dataflow](https://cloud.google.com/dataflow) (<https://cloud.google.com/dataflow>) to stream and batch data, [Cloud Storage](https://cloud.google.com/storage#section-1) (<https://cloud.google.com/storage#section-1>) to store log data backups, and [Looker](https://cloud.google.com/looker) (<https://cloud.google.com/looker>) to realize data visualization.

By cross-analyzing the user behavior data of its platform and users' feedback, the 17LIVE team is able to identify the features that attract its users and concentrate its development resources on upgrading these popular features to reach higher user satisfaction. When undesirable events like revenue drop or user loss occur, the team can quickly identify causes through data analytics on BigQuery and take timely actions.

Hsu notes that the seamless integration between different data tools of Google Cloud has significantly reduced the engineering effort required for data analytics. To further increase its data processing efficiency, 17LIVE leverages [Cloud Composer](https://cloud.google.com/composer) (<https://cloud.google.com/composer>) to schedule its data transferring workflows between Cloud Storage and BigQuery. This way, the team can easily perform daily analyses of app traffic and content performance, providing live streamers on its

streaming platform with suggestions to improve their viewer engagement, such as the best time to start live streaming.

To maximize the value of its data, 17LIVE has recently incorporated artificial intelligence (AI) into its data analytics. The company chose to build machine learning (ML) models in Vertex AI (<https://cloud.google.com/vertex-ai/>) using historical data in BigQuery, because Vertex AI supports simpler deployment compared to other similar offerings. The ML models in Vertex AI can now predict key contributors of each live streamer on 17LIVE's platform at an accuracy rate of 70%. With this data, live streamers can identify their fans more easily and have more engaging interactions with their viewers.

"The extensive data product range of Google Cloud has enabled us to fully unlock the value of our data without spending too many resources on system integration and management," says Hsu. "With the data insights generated by BigQuery and highly accurate predictions produced by Vertex AI, we know our users' preferences better and can help live streamers optimize the performance of their content."



Improving service stability with Vertex AI and Cloud Monitoring

Vertex AI also assists 17LIVE in troubleshooting by predicting potential system errors using the company's server log data. The 17LIVE operations team can now receive an alert 30 minutes before a system error happens and fix the issue immediately. Since the launch of this AI-powered error prevention

mechanism, 17LIVE has seen 20% less user reports complaining about system errors, and the number of streaming-related issues has decreased by 40%.

Meanwhile, the digital media company leverages Cloud Monitoring (<https://cloud.google.com/monitoring>) to monitor its microservices in GKE. Since all the log data can be easily transferred from GKE to Cloud Monitoring without passing any intermediate, 17LIVE can monitor the performance of its streaming platform in real time and troubleshoot in a timely manner. The stability of its streaming service is therefore further ensured.

"With the high-performance infrastructure and data analytics tools of Google Cloud, we've successfully enhanced the quality and variety of our live video streaming services. Next, we're looking forward to collaborating with Google Cloud to realize more innovative and interesting ideas in our streaming rooms," says Hsu.

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